Worked solutions — class questions and practice

Only consult these solutions after completing the relevant classwork. Check and correct your own reasoning. Homework tasks and homework answers are not included.

Part A — Desmos

1–2. Use points (mean time in s, position in m). The resulting x–t trend should be smooth and plausible; the model does not have to pass through every noisy measurement.

3–4. Use the gradient of a tangent or narrow secant to estimate velocity, stating the chosen time and units (m/s). Compare early and late values with your hand graph.

Part B — interval velocities

5. For each consecutive 5 cm interval, equation $v_{avg}=\Delta x/\Delta t$. Rearrangement: none. Substitute $\Delta x=0.05$ m and the difference in mean crossing times. Report m/s. Midpoint time = (earlier mean time + later mean time)/2.

6. Plot interval-average velocity (vertical, m/s) against midpoint time (horizontal, s). Each point approximates the velocity near that midpoint.

7. An approximately straight, upward-sloping v–t trend suggests near-constant positive acceleration. Scatter may limit how confidently this can be concluded.

8. Change in velocity per distance: $\Delta v/\Delta x$ (s^{-1}). Change per time: $\Delta v/\Delta t$ (m/s^2), which is acceleration. They are different quantities.

9. Accept any specific evidence-based limitation and feasible improvement with reasoning. Examples include release consistency, timing uncertainty and mark-reading accuracy.